

How Stale Can an Optical Digital Twin Be?

A Measured Freshness Budget for QoT Decisions

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Abstract: A GNPY-class digital twin makes format decisions between telemetry syncs while the network fills. We measure when its decisions fail: false-clears grow linearly with un-synced channel adds, giving an eight-add freshness budget.

1 Introduction

Digital twins of the optical physical layer are moving from research to practice [2]. A twin holds a QoT model of the deployed network, and operators use it to answer provisioning questions: which format will this lightpath sustain, how much margin does it need. The twin’s answer is only as good as its view of the network, and the view goes stale. Between telemetry syncs, channels are added, nonlinear interference grows, and the GSNR the twin believes is no longer the GSNR the receiver sees. The question every deployment must answer is how often the twin needs to sync. Today that answer is set by engineering judgment. This paper gives it a measured number.

We treat each twin decision as a certificate. The twin clears a format for a lightpath, and the certificate fails when the deployed channel cannot support that format. We call this failure a false-clear, and we measure its rate on the GN-model physical layer of GNPY [1] over the CORONET-CONUS reference network, with deployed traffic drawn from a real demand matrix. Four results follow. The false-clear rate of a stale twin grows linearly with the number of channels added since the last sync to the band a service shares along its path, which yields a freshness budget of about eight such adds at a 5% risk target. Which services are exposed when the twin is stale is set by spectral position and reach, so a per-lightpath view recovers real capacity over a uniform margin. With equal margin budgets, one service class does better under one allocation policy and the other class under the other, so a fleet-level number serves neither. And when the twin’s QoT model is a learned surrogate, training it on average error leaves it blind exactly at the FEC threshold where its decisions are made.

2 Method

Substrate. Lightpaths are transparent shortest-path routes of the CORONET-CONUS reference topology; per-channel GSNR comes from the GN model in GNPY [1]. Table 1 lists the parameters. The twin’s view is the band as last synced. The staleness variable N is the number of channels present in the

Table 1: Substrate parameters

Topology	CORONET-CONUS (75 nodes; routes ≤ 2500 km, 2–31 spans)
Fibre	SSMF, 0.2 dB/km, $D = 17$ ps/nm/km, 80 km spans
Amplifiers	EDFA, NF 6 dB
Launch power	+2 dBm per channel, flat (not re-optimised per fill)
Grid	32 GBd on 50 GHz, C-band of 76 channels
QoT engine	GNPY GN model (analytic NLI)
Deployed fill	sampled from SNDlib janos-us-ca per-link distribution
Estimator noise	0.3 dB monitoring; 0.5 dB RMSE in §3.2

C-band that co-propagates with the service along its own path (its interference neighbourhood), and ΔN counts the channels added to that band since the sync. N runs from a lightly filled provisioning band of 6 to a full grid of 76. A certificate selects the highest of five declared formats whose required GSNR the twin’s estimate clears: QPSK, 8QAM, 16QAM, 32QAM and 64QAM at 6.5, 9.0, 12.5, 16.0 and 19.0 dB respectively (32 GBd, roughly 20%-overhead soft-decision FEC; declared values, no probabilistic shaping). The capacity result of §3.2 uses a rate-flexible transceiver instead: spectral efficiency $\min(6, \log_2(1 + \text{SNR}/\Gamma))$ per polarisation with a $\Gamma = 1$ dB gap for FEC and implementation.

Certificate, consumer, witness. A certificate selects the highest format whose required GSNR the twin’s estimate clears. The consumer is that decision. The witness is the true deployed GSNR, in practice the receiver’s pre-FEC BER. For the surrogate study the features are the synced GSNR plus spectral position, reach and fill, the label carries an irreducible term for unmeasured network state, and two gradient-boosted objectives compete: average-error (squared loss) and decision-aware (a conservative quantile loss). Bars and manipulation checks are pre-registered. The sealed record is content-hashed and graded on seeds disjoint from those used during design [7].

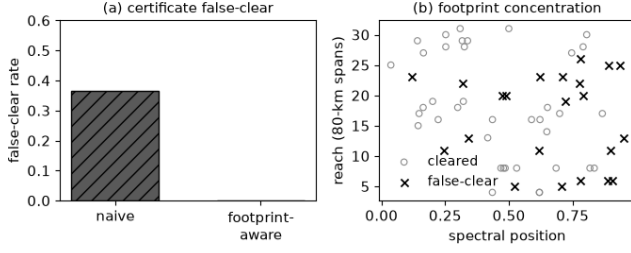


Figure 1: Where a stale twin fails (GNPy over CORONET-CONUS). Per-service false-clear over spectral position and reach concentrates on threshold-adjacent services. A per-lightpath view removes it. Marker shape distinguishes classes.

3 Results

3.1 The freshness budget

The risk of a stale twin is driven by the fill added since its last sync. Sweeping the deployed band from 6 to 76 channels while the twin’s view stays at the sync point, the false-clear rate of the twin’s format decisions rises linearly with the number of added channels, $FC \approx 0.008 \Delta N$ with $R^2 = 0.95$. At a 5% risk target the budget is about 8 added channels. The rule is concrete. Re-sync the twin, or re-certify the affected services, before the band around them grows by about 8 channels, rather than on a fixed calendar. Given an arrival rate the budget converts to time. A band taking one add per week can hold a sync for about two months. A busy one taking one add per day must sync roughly weekly.

3.2 Who is exposed when the twin is stale

Staleness does not spread its damage evenly. A lightpath’s GSNR drops as neighbours arrive, and whether it crosses a format threshold depends on where it sits: false-clears concentrate on services whose spectral position and reach leave their GSNR near a boundary, and are absent elsewhere (Fig. 1). So a single blanket margin cannot protect the fleet efficiently. We price this against today’s practice of one uniform system margin, with the deployed fill taken from real traffic. Routing the 1482 demands of the SNDlib janos-us-ca matrix [6] (39 nodes, 61 undirected links) gives a per-link channel-fill distribution with median 10 and 90th percentile 35 of 76; each CORONET lightpath’s deployed band fill is sampled from that distribution. Real traffic is heavy-tailed, so a uniform margin sized for the few near-full links over-provisions the lightly filled majority. Giving both policies the same estimator uncertainty of 0.5 dB RMSE, a twin that sizes each margin to the lightpath’s own fill recovers about 8% of spectral efficiency, near 0.33 b/sym or 21 Gb/s per channel, at matched safety (false-clear at most 0.01) on a rate-flexible transceiver model. The gain is capacity the uniform margin strands on links that never fill.

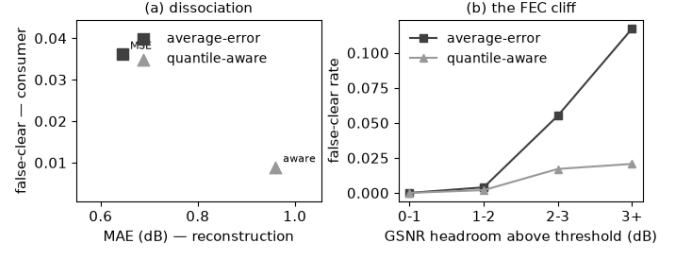


Figure 2: Two surrogate objectives, two metrics (GNPy, gradient-boosted regressors). (a) The average-error objective has lower MAE and higher false-clear than the quantile objective. (b) Its false-clears sit at the FEC cliff, where headroom is small. Marker shape distinguishes objectives.

3.3 One twin, two consumers, opposite verdicts

Exposure is not just uneven. It can invert. Take two margin policies with the same fleet-mean budget of $\bar{m} = 1.0$ dB. Policy A allocates by reach, $m_i^A = \bar{m} r_i / \bar{r}$ with r_i the service’s span count normalised to the longest route. Policy B allocates the same total by band-centrality, $m_i^B = \bar{m} c_i / \bar{c}$ with $c_i = 1 - 2|p_i - \frac{1}{2}| + 0.05$ for spectral position $p_i \in [0, 1]$. Long-reach band-edge services do better under A (false-clear 0.37 against 0.78), and short-reach band-centre services do better under B (0.22 against 0.42). Same pair of policies, opposite verdicts, on every seed tested. The fleet aggregate cannot arbitrate. It prefers A overall while B is much better for half the fleet. A control makes the mechanism sharp. We ran the same protocol on two FEC classes. Their thresholds differ by a shift, but they read the same projection of the network state, and no inversion appears. The inversion needs consumers that read different axes of the state, here reach against spectral position, and that is exactly what mixed service portfolios do. A twin that reports one fleet-level number answers a question no consumer asked.

3.4 The learned surrogate fails at the threshold

Many twins replace the physics model with a learned surrogate for speed. How that surrogate is trained matters more than how accurate it is on average. Uncertainty-aware and quantile QoT models are established [3]; what has not been measured is the price of ignoring them at the decision boundary. Under irreducible prediction uncertainty the average-error surrogate wins the metric it was trained on and loses the one the twin exists for. Its MAE is 0.65 dB against 0.95 dB for the quantile objective, yet its false-clear rate is about 0.041 against 0.008, five times higher, at comparable delivered spectral efficiency near 3.9 bits/symbol (Fig. 2a). The failures sit where the GSNR headroom above the nearest format threshold is small (Fig. 2b). That is the cliff, and a symmetric error metric is blind there. A twin’s surrogate should be trained on the decision it will make, and the quantile machinery to do so already exists [3].

4 Related work

Optical digital twins are an active area, from GNPY-based twins fed by network telemetry [2] to calibration against field data; that work builds and tunes the twin, and does not quantify how stale it can afford to be. Margin erosion under channel add is known qualitatively, and impairment-aware provisioning accounts for it at setup time [5]; our contribution is the measured staleness law and the budget it implies for the twin that persists after setup. Uncertainty-aware ML-QoT, including quantile models, is established [3, 4]; we credit it as the correction for the surrogate result and add the measured cliff dissociation that motivates it. We do not offer a better twin, estimator or allocator. We offer the freshness budget, the exposure map, and the price of the wrong training objective.

5 Discussion

A digital twin is a certificate factory, and certificates age. The freshness budget turns twin synchronization from a judgment call into a provisioned quantity: re-sync within eight adds to a service's band at a 5% risk target, protect the threshold-adjacent services first, and train surrogates on the decision rather than the average. The evidence stands on the GN model with real demand data, and its limits are worth stating. Launch power is flat at +2 dBm and not re-optimised as the band fills, ROADM filtering penalties are not modelled, and the witness is the GN-model GSNR rather than a receiver's pre-FEC BER. Each of these moves deployed GSNR further from the twin's synced view, so they tighten the budget rather than relax it. A coherent-receiver witness on a testbed twin is the natural next step. The same consumer-relative failure appears in radio link adaptation and in database replication [7]. The results here stand on the optical QoT certificate alone.

References

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